

Chapter 20: Geometric Graphs

We consider some several questions about general graphs drawn in the plane. When we restrict to straight-line segments for edges, these are sometimes called geometric graphs.

20.1 Crossing Number

We saw earlier that only relatively sparse graphs are planar. The question of genus is then an extension of planar. Another direction is to count the number of times two edges cross. The smallest value of this is called the *crossing number* of the graph. For example, it is not hard to show that the crossing number of both K_5 and $K_{3,3}$ is 1: if one removes one edge the result is planar, and one can add the missing edge and have it only cross one other edge.

The lower bound can be generalized to say that the crossing number is at least $E - (3V - 6)$, since each edge beyond $3V - 6$ must cross some edge. For example, it follows that K_6 has crossing number at least 3, and it can be drawn as such. But for dense graphs this bound is weak, since most edges will cross several edges. The crossing number of the complete graph is unresolved. It is known that:

THEOREM. *The crossing number of the complete graph lies between $\frac{1}{80}n^4$ and $\frac{1}{64}n^4 + O(n^3)$.*

20.2 Geometric Graphs

One can now consider graph questions as they relate to the geometry. For example, one might ask for the minimum number of edges that ensure one can find two edges that do not intersect. This was resolved by Erdős and others. We assume no three vertices are colinear.

THEOREM. *If a geometric graph with more edges than vertices, then there exists two disjoint edges that do not cross.*

PROOF. If there is a vertex of degree 1, then remove the vertex and its edge, and apply induction. So assume the minimum degree is at least 2. Let v be a vertex with degree more than 2, say with neighbors w_1, w_2, w_3 . If w_1, w_2, w_3 all lie in some half-plane created by a line through v , then let w_2 be the “middle” neighbor looking from v . Otherwise let w_2 be any neighbor. Then the second edge incident with w_2 cannot intersect both vw_1 and vw_3 . This gives us the desired pair of edges. $\square$

Then one can try to improve this to a set of k edges that form a matching in the graph and are non-intersecting. After several authors provided results for small k , Pach and Törőcsik used Dilworth’s theorem to show that a linear number of edges suffice.

Another direction is to consider that we have all edges, that is a drawing of the complete graph, and then the question is what size of non-intersecting or pairwise-intersecting matching is guaranteed. As before we require that no three vertices are colinear.

The answer to the first is perhaps surprising: one can find a (almost)-perfect matching none of whose edges intersect. The proof is to take the maximum matching with the shortest total length! If two of its edges cross, say ab intersects cd , then by the triangle inequality, one can replace this pair by either ac and bd , or by ad and bc , and so produce a shorter total length, a contradiction.

One can invert the second question and ask, how many vertices do we need to produce a crossing family of size k . For example, it is not hard to show that any drawing of K_5 must have a matching of size 2 whose edges intersect. This problem is related to the question of how many points does one need to ensure that there is a convex quadrilateral. In general, exponentially many vertices are needed.